

Dr. ALLWIN SUDHAKARAN, M.Sc., Ph.D., Post-Doc.

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PROFILE

Dedicated Physicist specializing in material science with expertise in electrocatalysis (HER, OER, ORR) and CO₂ utilization. Combines hands-on experimental mastery of metal/metal oxides and nanocomposites with Density Functional Theory (DFT) studies to elucidate reaction mechanisms and electronic properties. Passionate about research and education, with a strong record of utilizing advanced scientific instrumentation and computational tools to drive discovery. Published extensively in high-impact journals.

EDUCATION

- ❖ **Research Scientific Staff (2025-Present)** – IISER Thiruvananthapuram, Kerala, India.
- ❖ **Postdoctoral Fellow (2023-2025)** – Jain University, Karnataka, India.
- ❖ **Ph.D. in Physics (Material Science) (2023)** – Karpagam University, Tamil Nadu, India.
- ❖ **M.Sc. Physics (95.4%) (2019)** – Karpagam University, Tamil Nadu, India.
- ❖ **B.Sc. Physics (87.4%) (2017)** – Karpagam University, Tamil Nadu, India.

WORK EXPERIENCE & INTERNSHIPS

- ✓ **Research Intern** – Indira Gandhi Centre for Atomic Research (IGCAR), DAE, Kalpakkam (APR 2018 to JUL 2018).
- ✓ **M.Sc. Final Year Project Trainee** – IGCAR, DAE, Kalpakkam (SEP 2018 to APR 2019).
- ✓ **Reviewer** for Springer Nature & Elsevier Journals (2024-Present)

RESEARCH & EXPERTISE

- ✓ DFT and Experimental Electrocatalysis, CO₂ reduction & fixation, HER, OER, and ORR reactions.
- ✓ Synthesis & characterization of metal/metal oxides, nanocomposites, ferrites, mixed metals, multifunctional materials, and thin films.
- ✓ Structural, microstructural, morphology, elasticity, vibrational analysis, Magnetic nanostructures, memory devices, and optoelectronic applications.

✓ Instruments handled:

CH instrument, Scanning Probe Microscopy (SPM: AFM and MFM Mode), Differential Scanning Calorimetry (DSC), Rotary evaporator, Thermal Vacuum Evaporation coating unit (Thin Film coating), Thermo Gravimetric Analysis (TGA), Zeta potential, Fourier Transform Infrared Spectroscopy (FTIR), Electrospinning, Dynamic light

scattering (DLS), Ultraviolet-visible Spectroscopy (UV-Vis), Photoluminescence (PL).

PUBLICATIONS & CONFERENCES

- 30+ International Journal Publications (SCI, Scopus Indexed) and 01 patent (filed).
- Published in Progress in Materials Science (IF: 40.8), Catalysis Reviews (IF: 13.6), Sustainable Materials and Technologies (IF: 8.7), Fuel (IF: 6.7), etc.
- 30+ International Conferences & Presentations, 7 workshops
- Published 09 International Books.

TECHNICAL & SCIENTIFIC SKILLS

❖ Software used:

- **Structure Refinement:** Full-Prof, MAUD, GSAS-II, JANA 2006, HighScore Plus, Match-3, Profex, Pcpdfwin, and TOPAS.
- **Bond Diagrams:** VESTA, Avagadro, ChemDraw Ultra, Ortep, and WInGX.
- **FTIR Analysis:** OMNIC, VINUS, and IRPal.
- **Graph Plotting:** Origin Lab, MAT LAB, and QtiPlot.
- **DFT Calculations:** Gaussian, Quantum Espresso, Material Studio, Wein2k, Orca, COMSOL, OOMMF.
- **Plagiarism:** Turnitin, etc.

AWARDS & MEMBERSHIPS

- ✓ Best Research Paper Award, International Conference (2023)
- ✓ Member of IAENG, IFERP, AICTSD
- ✓ Guinness World Record Participant (2010) – Largest One-Day Art Competition.

LANGUAGES

English | Tamil | Hindi

REFERENCES

Dr. A. Saranya, Assistant Professor, Vel Tech Rangarajan Dr. Sangunthala R&D Institute of Science and Technology, Chennai-600062, Tamil Nadu, India (saran.amirth@gmail.com)

Dr. G. Mangamma, Associate Professor, Homi Baba National Institute (HBNI) & Scientific Officer H, Indira Gandhi Centre for Atomic Research, Kalpakkam 603102, Tamil Nadu, India (gm@igcar.gov.in)

Dr. P. Pandaram, Scientific Officer H, TLD Lab, Kundakulam Nuclear Power project, Kundakulam, Tamil Nadu-627106, India (dr.p.pandaram@gmail.com)

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